

Education

- **University of Wisconsin-Madison** Madison, WI
4.0 GPA, Computer Science Major, Starting Grad School Fall 2027 September 2024 - May 2027
 - Taken: Deep Learning for NLP, Intro to Algorithms, Machine Organization and Programming, Intro to AI, Intro to Computer Engineering, Intro to Discrete Mathematics
- **McNeil High School** Austin, TX
5.4 weighted GPA, 1510 SAT 2020 - 2024
 - Enrolled in: Physics C, Calculus BC, Mobile Applications, and Digital Electronics

Relevant Experience

- **Extreme Engineering Solutions (X-ES)** Verona, WI
Embedded Software Engineering Co-op May 2026 - December 2026
 - Authored Linux and VxWorks device trees for two new rugged embedded boards by cross-referencing hardware schematics, processor manuals, and device trees from similar existing boards.
 - Used those device trees to build and validate internal Buildroot-based Linux releases for both boards, integrating board-support configurations and testing extensively.
 - Produced customer-facing Gentoo Linux releases for both boards by debugging a legacy build pipeline, fixing broken API calls and migrating scripts from Python 2 to Python 3; authored an internal guide documenting the process for future engineers.
 - Additionally, validated a new release of X-ES Enterprise Linux (XEL), a RHEL-based OS distribution, by running the integration test suite and manually verifying command output and system logs against expected results, then updated the corresponding product manual.
- **Wisconsin Autonomous Vehicle Club** Madison, WI
Perception Lead January 2025 - present
 - Rewrote lane segmentation (drivable area detection) for our autonomous car's perception stack in C++ using TensorRT; combined with teammates' object detection and lane line rewrites, contributed to a 142% increase in framerate.
 - Leading a 10-20 member perception team building the vehicle's real-time computer vision pipeline using 3D stereo cameras.
 - The team placed 2nd of 10 in AutoDrive II for the past two years.
- **Undergraduate TA** Madison, WI
CS540: Intro to AI Spring Semester 2026
 - Held weekly office hours, helping students work through homework assignments and understand topics ranging from linear regression and unsupervised learning to transformers and reinforcement learning.
- **Internship with the Institute for Computing in Research** Austin, TX
Intern July 2024
 - Created a neural network to play chess under the direction of Los Alamos scientist Mark Galassi.
 - Code: <https://github.com/asventon1/chess2> Paper: <https://tinyurl.com/3spr2vk7>
- **Personal Projects**
My own creations (with videos for some and code for all) 2017 - present
 - Google street view mesh generator: <https://github.com/asventon1/goondaMaps>
Generates a 3D point cloud from a Google Street View panorama.
 - Chess engine: <https://github.com/asventon1/chess>
Engine uses the minimax algorithm for playing chess and Java's graphics engine for rendering.
 - Stock prediction: <https://tinyurl.com/yjunk43r>
Experiment using Google agentic AI to create an app to predict future stock prices.
 - Fingers: <https://github.com/asventon1/fingers>
Uses TensorFlow and a convolutional neural network to detect the number of fingers in a hand gesture.

- HEB slot getter: <https://github.com/asventon1/slot-getter>
Checks the HEB grocery store website for curbside slots and notifies you if there is an open one.
- Hamster tracker: <https://github.com/asventon1/oreo>
Makes use of an Arduino and proximity sensors to track and graph the activity of a hamster wheel.
- Metaballs: <https://github.com/asventon1/metaballs>
Experiment with rendering metaballs.

- **Computer Science UIL**

Austin, TX

- *Participant*

2022 - 2023

- 3rd place in the individual written portion and 2nd place in the team programming portion.

Qualifications

- Languages: Python, C, C++, Java, JavaScript, Bash
- Embedded & Systems: Linux (Buildroot, Gentoo, RHEL), VxWorks, Device Trees, Serial Console, Arduino
- Tools: Git, Make, CMake
- ML/Computer Vision: NumPy, TensorFlow, PyTorch, TensorRT

Writing

- Here are some of my college essays if you want to see some of my writing
<https://drive.google.com/drive/folders/165P2t10FTcPWZYHtIjUyIKpu1tL4bHOP?usp=sharing>